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Hiwonder

Glowing Ultrasonic Sensor User Manual

V1.0

1. Glowing Ultrasonic Sensor Instruction

1.1 Sensor Introduction

This is a glowing ultrasonic ranging module with an industrial-grade ultrasonic ranging chip, where the chip integrates an ultrasonic transmitter circuit, ultrasonic receiver circuit, digital processing circuit, etc. The module adopts I2C communication interface, which can read the range measured by ultrasonic sensor through I2C communication.

1.2 Working Principle

The working principle of distance measurement is that the module will automatically transmit 8 square waves at 40khz during ranging, and then detect whether there is a signal return.

Formula: measurement distance=(high level time*sound speed (340m/s))/2.

Two RGB lights are integrated into the ultrasonic probe.It's not only can adjust the brightness of the light, but also change three color channels of red(R), green(G), and blue(B) and superimpose them on each other to transfer colors between ultrasonic waves.

This module is widely used, mainly applied to realize automatic avoidance function of intelligent vehicles and intelligent robots.

2. Caution

Please don't use glowing ultrasonic sensors beyond the rated voltage range.

3. Parameters Instruction

For information about the chip in this module, please refer to “3. Chip Manual and Schematic Diagram”.

3.1 PIN Instruction

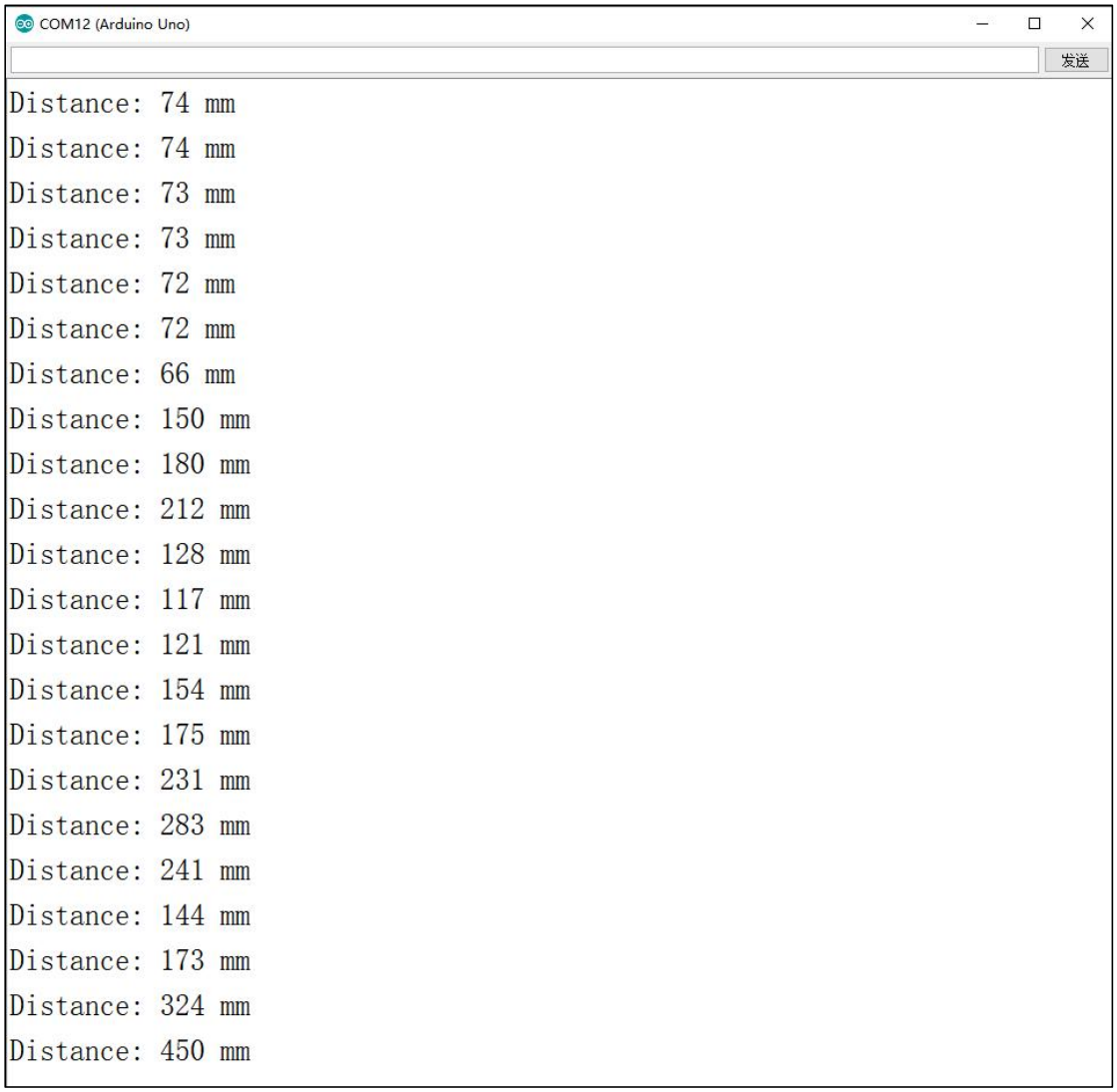
PIN	PIN Instruction
5V	Power Input
GND	Power ground
SDA	SDA Cable
SCL	SCL Clock Line

3.2 Specification Parameters

Ultrasonic Sensor	
Prameter Name	Specification
Supply Voltage	DC 5V

Working Current	2mA
Working Frequency	40Hz
Detection Distance	2cm~400cm
Measurement Angle	15°
Communication	IIC
I2C Address (7 bits)	0x77
Port	5264-4AW
Working Temperature	0℃~50℃
Size	46mmx20mm
Modular installation, compatible with LEGO series	

4. Program Outcome

You can go to the same directory in this section to view case tutorials and programs for various platforms. In this article, we only use Arduino IDE as an example to demonstrate the detection results.

The distance of the obstacles in front is detected by using glowing ultrasonic sensor. At the beginning, we obstruct the object in front of the sensor, then slowly increase the distance. At this time ,you can see the values displayed in the serial monitor are continuously increasing. In contrast, when we bring the obstacle closer, the values displayed in the serial monitor are decreasing. And during this process, two RGB lights are used to indicate the distance of obstacles. When the distance is too close, it will display red; when the distance is too far, it will display green.